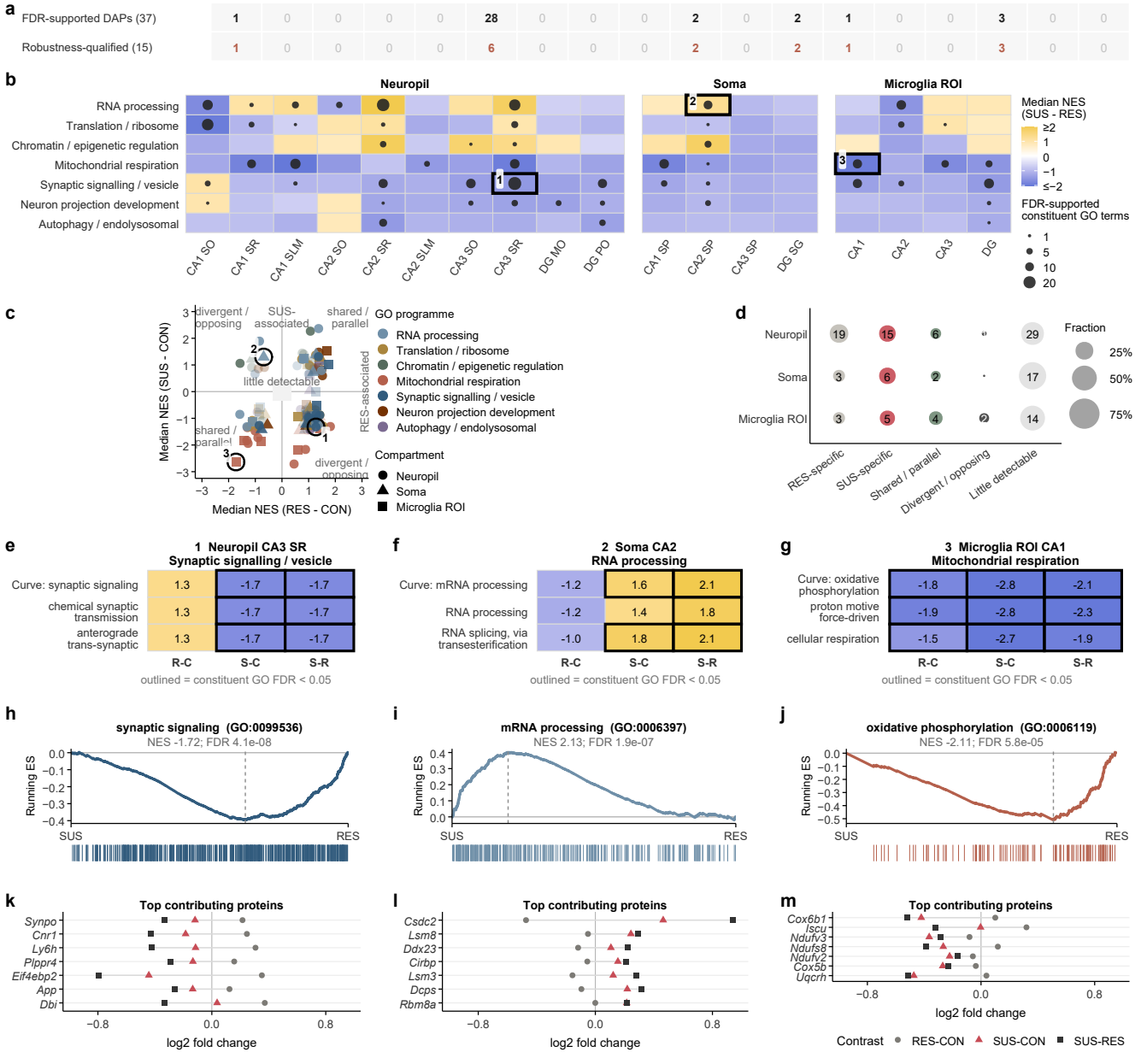

a
FDR-supported DAPs (37)
1 0 0 0 0 28 0 0 0 2 0 2 1 0 0 3 0 0
Robustness-qualified (15)
1 0 0 0 0 6 0 0 0 2 0 2 1 0 0 3 0 0
b
Neuropil
Soma
Microglia ROI
RNA processing
Translation / ribosome
Chromatin / epigenetic regulation
Mitochondrial respiration
Synaptic signalling / vesicle
Neuron projection development
Autophagy / endolysosomal
CA1 SO
CA1 SR
CA1 SLM
CA2 SO
CA2 SR
CA2 SLM
CA3 SO
CA3 SR
DG MO
DG PO
CA1 SP
CA2 SP
CA3 SP
DG SG
CA1
CA2
CA3
DG
Median NES (SUS - RES)
≥2
1
0
−1
≤−2
FDR-supported constituent GO terms
1
5
10
20
c
Median NES (SUS - CON)
Median NES (RES - CON)
divergent / opposing
SUS-associated
shared / parallel
little detectable
RES-associated
GO programme
RNA processing
Translation / ribosome
Chromatin / epigenetic regulation
Mitochondrial respiration
Synaptic signalling / vesicle
Neuron projection development
Autophagy / endolysosomal
Compartment
Neuropil
Soma
Microglia ROI
d
Neuropil
Soma
Microglia ROI
19 15 6 29
3 6 2 17
3 5 4 14
Fraction
25%
50%
75%
RES-specific
SUS-specific
Shared / parallel
Divergent / opposing
Little detectable
e
1 Neuropil CA3 SR
Synaptic signalling / vesicle
Curve: synaptic signaling
chemical synaptic transmission
anterograde trans-synaptic
1.3 -1.7 -1.7
1.3 -1.7 -1.7
1.3 -1.7 -1.7
R-C S-C S-R
outlined = constituent GO FDR < 0.05
f
2 Soma CA2
RNA processing
Curve: mRNA processing
RNA processing
RNA splicing, via transesterification
-1.2 1.6 2.1
-1.2 1.4 1.8
-1.0 1.8 2.1
R-C S-C S-R
outlined = constituent GO FDR < 0.05
g
3 Microglia ROI CA1
Mitochondrial respiration
Curve: oxidative phosphorylation
proton motive force-driven
cellular respiration
-1.8 -2.8 -2.1
-1.9 -2.8 -2.3
-1.5 -2.7 -1.9
R-C S-C S-R
outlined = constituent GO FDR < 0.05
h
synaptic signaling (GO:0099536)
NES -1.72; FDR 4.1e-08
Running ES
SUS
RES
i
mRNA processing (GO:0006397)
NES 2.13; FDR 1.9e-07
Running ES
SUS
RES
j
oxidative phosphorylation (GO:0006119)
NES -2.11; FDR 5.8e-05
Running ES
SUS
RES
k
Top contributing proteins
Synpo
Cnr1
Ly6h
Plppr4
Eif4ebp2
App
Dbi
log2 fold change
l
Top contributing proteins
Csdc2
Lsm8
Ddx23
Cirbp
Lsm3
Dcps
Rbm8a
log2 fold change
m
Top contributing proteins
Cox6b1
Iscu
Ndufv3
Ndufs8
Ndufv2
Cox5b
Uqcrh
log2 fold change
Contrast
RES-CON
SUS-CON
SUS-RES